Technical Paper

The Philips/Burle Full Height Turnstile Controller & Electronics
(more than you ever wanted or needed to know)

The original STG series of full height turnstiles were manufactured by Burle Industries. Burle turnstile products were a step up from the products they acquired when they purchased a company known as Robot Industries. The Burle product employed an electronic controller that was originally designed to operate motorized gates. Subsequently, Burle was acquired by Philips Communication & Security Systems (Phillips CSS). Philips redesigned the electronics and employed a mini Programmable Logic Controller (PLC). Philips then became Dearborn Access Control just prior to being absorbed by Tomsed. While the design was functional, it had several fatal design flaws. It is important to keep in mind that these issues are strictly electrical. The turnstiles are mechanically superior in every respect.

1- The Philips PLC version of the controller utilizes a 24VAC, 6.4 amp transformer with a full wave bridge rectifier and no DC filtering. All transformers, being non-linear devices, have some degree of voltage regulation, provided they have a load connected to them. The rule of thumb is that the resultant full wave DC voltage is 1.4 times the AC voltage. For a transformer that outputs 24VAC exactly, the resultant DC voltage would be 33.6VDC. The transformer provided by Burle & Philips was grossly under-loaded and could produce AC voltages exceeding 27VAC resulting in possible voltages of 37.8VDC or greater. How the transformer was loaded could depend on the particular configuration of Fail-lock and Fail-safe solenoids. This excessive voltage resulted in premature solenoid failure as well as PLC failure.

2- The original Philips solenoid was designed with dual coils and an internal/integral limit switch. Upon initial application of power, the high power coil would be energized. When the solenoid plunger reached its seated position, the limit switch would activate the low power holding coil. Over a period of time this internal limit switch would take a beating and fail. This resulted in prolonged operation of the high power coil and consequently solenoid failure. (The new replacement solenoids that we offer are electronically controlled and do not have this limit switch problem).

3- The PLC is a microprocessor based devices with operating limits of 24VDC plus or minus 10%. That translates to 21.6 to 26.4 VDC operating range. Obviously, way off the chart given the power supply output. In addition, the unfiltered DC is unfriendly to microelectronics. Furthermore, Philips designed the PLC to directly switch the solenoids on and off via the marginally sufficient 2 amp rated internal PLC relays without proper arc suppression. Solenoids are electrically inductive devices (coils). They characteristically produce a spike in voltage when de-energized (arcing). This produces pitting on the relay contacts that eventually causes them to malfunction.

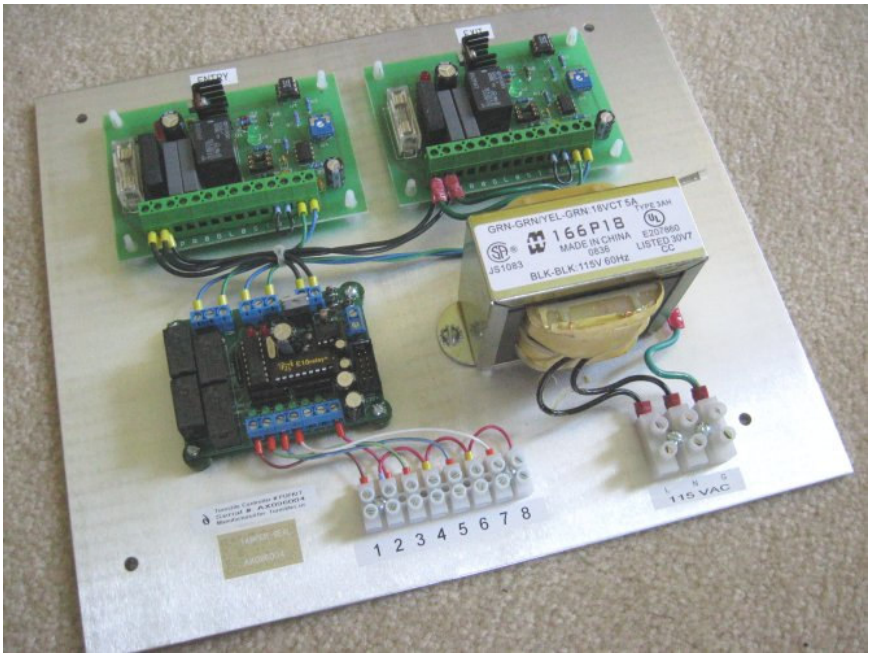
4- During the redesign from a Burle product to a Philips product the override function was eliminated.

Tomsed purchased the turnstile division from Philips and continued to support the legacy Philips and Burle products in the field. Tomsed continued to produce new STG turnstiles with its own propriety mechanism and control electronics. As a corrective measure for legacy Burle & Philips products, Tomsed initially approached the problem of excessive voltage by providing a loading resistor for the transformer. Although this helped reduce the failures, it did not address all known issues. Subsequently, Tomsed came out with its Version 2 Upgrade. That solution replaced the transformer, bridge rectifier and Philips firmware based PLC with all new electronics including new PLC firmware.

Subsequently, Tomsed was acquired by Boon Edam who has now dropped all support for legacy Robot, Burle and Philips products. Customers needing support for these turnstiles are offered an expensive top channel upgrade instead. This includes an all new mechanism and control electronics. Whether you need to replace the mechanism or not, that is all you are offered if you problem is only the electronics.

As you can see, we understand the problem and its history. We understand that it is wise to maximize your initial investment in your existing security system in these fiscal times and that it is a waste of resources to throw away your perfectly good and robust turnstile mechanism. We can help you extend the useful life of your turnstile.

We offer an affordable approach to supporting your Philips, Burle and Robot access control turnstiles. Our model PUPKIT is a PLC based controller with all new firmware and supporting electronics that also restores the override function. It addresses and corrects all known prior issues and limitation of its predecessors. Nobody understands the problem better than we do and nobody is equipped to help you better.



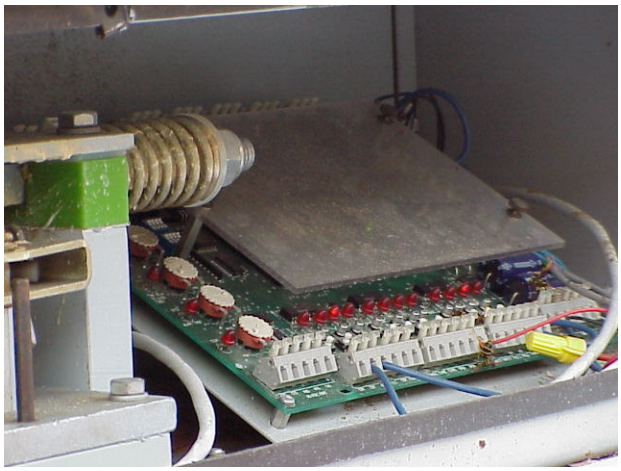
PUPKIT Replacement for Robot, Burle & Philips Full Height Turnstiles

IDENTIFICATION



Robot Industries Control Relay

Obsolete part numbers include: 53-521 with the name Dayton or Potter-Brumfield



Burle Gate Controller aka Multicontroller

Obsolete part numbers include: 51-101, 70-992



Philips & Tomsed PLC Controllers

The PLC label will indicate either Aromat, Nais or Panasonic FP0-C10RS

Other markings will show STGXXX, 8.00 or 8.03

Obsolete part numbers include: 73-102, 51-215, 86-289, 86-292